Kumar Chakraborty

689-242-4947 | kumarchakraborty3007@gmail.com | linkedin.com/in/kumarchak/ | github.com/kumarchak30

Education

University of Florida — **Expected May 2029**
B.S. Computer Engineering — *GPA: 3.49/4.00*

Technical Skills

Languages: Java, Python, C++, HTML/CSS/JavaScript, VHDL
Developer Tools: Git, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse, CLion, Intel Quartus Prime, Terasic DE-10 Lite FPGA
Libraries: NumPy, Matplotlib, Seaborn, SFML, Open CV, MediaPipe, Pyserial
Certifications: CompTIA IT Fundamentals Pro
Courses: EEL3701C Digital Logic and Computer Systems, COP3502C/COP3503C Programming Fundamentals 1 and 2

Experience and Involvements

Technical Affairs Director — **Apr. 2026 – Present**
Freshman Leadership Engineering Group (FLEG) at UF — *Gainesville, FL*

- Selected from a pool of 250+ candidates for standout leadership skills.
- Collaborated with a committee of 10 engineering students to plan and execute 3+ technical development events per semester, increasing underclassmen engagement through hands-on workshops.

Data Acquisition — **Jan. 2026 – Present**
Swamp Launch Rocket Team at UF — *Gainesville, FL*

- Redesigned PCB connections between pressure gauges, load cells, and the onboard data acquisition unit to improve data reliability during static and flight testing.
- Implemented Python scripts to process sensor data and deliver structured datasets to the physics team for thrust and pressure analysis.

Open Project Space (OPS) Member — **Sep. 2025 – May 2026**
Institute of Electrical and Electronics Engineers (IEEE) at UF — *Gainesville, FL*

- Collaborated in year-long curriculum of hardware and **Arduino** programming lectures.
- Beginner program to teach circuitry and hardware essentials to freshman Computer/Electrcial Engineering students.

Software Development Intern — **May 2024 – Jul. 2024**
GDN InfoTech — *Indianapolis, IN*

- Developed a Database Management system to automate the finances of local family-owned businesses using **Tkinter** and **Reportlab** in **Python**.
- Implemented core database functionalities, including query execution, transaction logging, and data visualization through a user-friendly interface.
- Collaborated in a startup environment, contributing to system design, programming, and testing while ensuring scalability and ease of access for small-business clients.

Projects

Minesweeper | *COP3503C Programming Fundamentals 2* — **Nov. 2025 – Dec 2025**

- Built a complete C++ Minesweeper implementation using SFML, applying object-oriented design principles, event-driven programming, and modular architecture.
- Implemented recursive reveal algorithms, configuration-driven board initialization, and real-time state tracking for gameplay logic and UI synchronization.
- Developed custom managers for textures and sprites to improve resource handling and maintainability, leveraging STL containers and RAII principles.
- Integrated persistent storage for player statistics using file I/O, along with debugging and visualization tools to support testing and validation.